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Seasonality defines groundwater recharge thresholds across Australia

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Keywords: groundwater recharge, fractured rock, rainfall recharge threshold, seasonality

Key Points:

- 300 potential groundwater recharge events from 16 locations across Australia (2022-2025) identified.
- In water limited environments (evapotranspiration dominating over rainfall) episodic rainfall events generate potential recharge almost exclusively during wet seasons
- A conceptual model is developed for progressive seasonal recharge response to rainfall events.
- Sustainable groundwater management should be informed by wet season rainfall events over the median rainfall recharge threshold.

Abstract

Reliable prediction of groundwater recharge from rainfall is crucial to water management but is scientifically challenging. Previous studies have demonstrated that caves, mines, and tunnels provide useful “windows” for observing rainfall recharge timing and thresholds (i.e., amount of rainfall in a given time period causing a change in dripping from the void ceiling). Using drip loggers at sites across the Australian continent, we find that annual median rainfall recharge thresholds are statistically similar across 16 sites. However, thresholds vary by season. In climates dominated by winter rainfall, recharge is highly unlikely to occur during summer and autumn, even for high volume rain events. Similarly, during drought years winter rainfall may not be sufficient to cause recharge, even for events where rainfall recharge thresholds are met. Conversely, montane and summer-wet sites showed a more reliable recharge response to rain events above the rainfall recharge threshold. This enhanced understanding of recharge dynamics should be considered in modelling aquifer dynamics and planning sustainable groundwater use.

1 Introduction

Groundwater is an important freshwater resource for humans, especially in arid regions. Growing demands on groundwater supplies has led to global overdraft of aquifers, with many aquifers experiencing significant declines in water levels (Famiglietti, 2014; Rodell et al., 2018; Wada et al., 2010). Anthropogenic climate change has been identified as a primary stressor impacting global groundwater resources, as alterations in rainfall patterns, soil temperature and moisture, and evapotranspiration increasingly change recharge patterns (Green et al., 2011; McCallum et al., 2010). Therefore, sustainable use of groundwater resources into the future requires management informed by an accurate understanding and quantification of the recharge processes that replenish our water supplies. This includes knowledge of rainfall recharge thresholds (the volume of rainfall required over a given period to induce groundwater recharge) and how that threshold is linked to climate, seasonality, geology, vegetation, and other factors.

There is an increasing number of studies examining how climate change will impact groundwater recharge at the global scale (Berghuijs et al., 2022, 2024). However, this analysis is still impeded by high uncertainty, with different models producing disparate projections of impacts (Refsgaard et al., 2016). In semi-arid to arid regions, where groundwater recharge volumes are low, this discrepancy is particularly apparent. Given the inherent link between terrestrial water cycles and climatic drivers, resolution of the predictions of impact in these regions therefore requires a better understanding of how and when rainfall events lead to groundwater recharge. Australia, a continent dominated by semi-arid and arid climates and with high availability of both climate and hydrological data, provides a good setting to further explore this relationship. Although several previous studies, both continental (Crosbie et al., 2013; Lee et al., 2024) and global (Berghuijs et al., 2022; Jasechko et al., 2014; Moeck et al., 2020) have estimated and modeled Australian groundwater recharge patterns, they have considered long-term patterns in groundwater recharge, ignoring the event-scale and inter-annual processes. However, as with many semi-arid regions, seasonality of rainfall is a

particularly important hydrologic driver that warrants further examination as a control on the occurrence of recharge.

In this study, we take advantage of Australia’s National Groundwater Recharge Observing Network (NGROS), introduced by Baker et al. (2024), to further explore the role of seasonality in determining links between rainfall and groundwater recharge. Using the first year of NGROS data, Priestley et al. (2025) provided an estimate of the amount of precipitation needed to generate recharge as a proof of concept at eight sites. We build on that study, using up to three years of field data across 16 sites, with a refined focus on seasonal variation in rainfall recharge estimates across a wide area of the continent. We then develop a conceptual model that provides the process understanding of rainfall-recharge links in seasonally dominated rainfall regions.

2 Materials and Methods

In this work we measure the onset of one-dimensional, direct infiltration into the cave, causing a change in the drip rate, which we henceforth refer to as ‘Potential Recharge’, acknowledging that we do not directly observe a change to the groundwater table.

2.1 Study sites

Between September 2022 and August 2025, drip data was collected at 16 sites in five Australian states (Figure 1). The project employed commercially available Stalagmate © loggers (Gemini Data Loggers UK Ltd). They consist of a pressure transducer in a fist-sized, sealed PVC Tinytag© enclosure to detect water dripping onto their surface. In-built hardware detects drip impacts vs non-drip noise (e.g., vibrations). The loggers were placed at 3 – 9 locations within each site and programmed to record drips directly onto the surface of the unit in 0.25 -- 1 hr intervals (Baker et al., 2024). The sites include natural caves, tunnels, and mine adits located a range of climate, geology, and vegetation (SI tables 1 and 2). All loggers are located at least 5 m below land surface to avoid the root zone as far as possible; while the root extinction depth of many eucalyptus species is much deeper, the high-density root zone associated with highest water uptake is generally within a couple meters of the soil surface (Knight, 1999). Thus, drips that penetrated more than 5 m were considered to represent potential groundwater recharge.

All sites except Capricorn Caves (CAP) and Wellington Phosphate Mine (WPM) are located within the temperate climate classification (Bureau of Meteorology, Government of Australia (BoM). WPM sits within the arid Steppe climate (Beck et al., 2023) (SI Table 2), while CAP is within the subtropical climate zone. All sites represent fractured rock aquifers; nine are in limestone (karst) regions, eight of which are caves and one is a mine. One cave is in volcanics. The remaining mines and tunnels are predominantly in consolidated sedimentary or metamorphosed sedimentary materials. Soil depth in the area of the sites has been estimated at 0.8 – 1.4 m (Viscarra Rossel et al., 2014), although observations in the field were that it was variable above the study sites, with exposed rock at some sites (Figure S1). Apart from CAP (dry rainforest), the official vegetation type at all sites is described as either eucalyptus woodland or eucalyptus forest, with variations on forest density, tree size, and understory vegetation (SI Table 2). However, the vegetation at most sites has been disturbed to some degree and is

sparse or non-existent at mine sites (SI Table 2 and Figure S1). According to the Bureau of Meteorology seasonality characterization (Figure 1 – Bureau of Meteorology, 2025), for 11 sites rainfall is predominantly in the winter and 3 sites have uniform rainfall throughout the year, although both border on winter rainfall areas. At Capricorn Caves and Willi Willi Caves (WIL) rainfall is predominantly in the summer months.

Average annual precipitation and modelled potential evapotranspiration (PET) was obtained from the Australian Water Outlook, which inputs gridded daily Bureau of Meteorology data from 1911 onwards into the Australian Resources Assessment Landscape model (<https://awo.bom.gov.au>, accessed 13 October 2025). The aridity index (AI) was then calculated as quotient of the average precipitation divided by the potential evapotranspiration, compared to Zomer et al. (2022) for validation, and determined to be representative of typical Australian conditions.

At least one year of data is available for each site, although the inclusion of multiple years of data at several sites allows for a more extensive temporal examination of rainfall recharge thresholds. At sites NAR, WIL, WAL, MON, BYA, and YAR (Fig 1) the data record has been extended to up to approximately 3 years, enabling additional analysis.

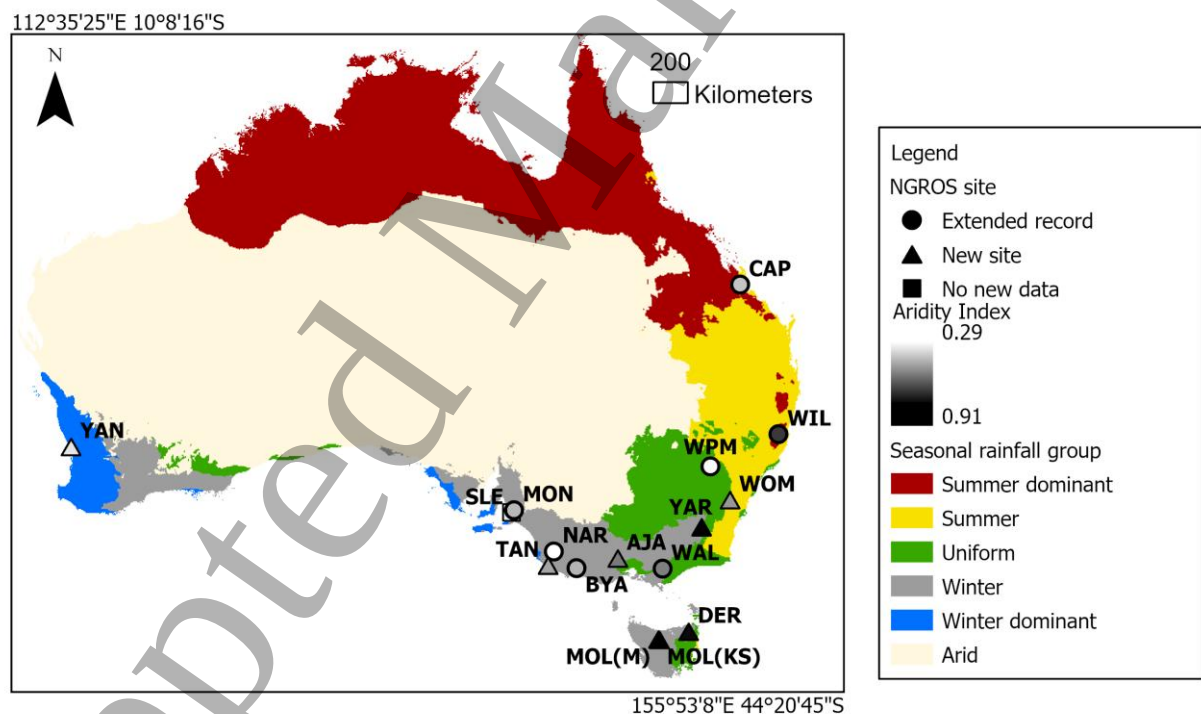


Figure 1. Map of the NGROS network in Australia. Site names and characteristics are further explained in SI Table 1. The network includes sites (shaded according to aridity index) that were analyzed in Priestley et al. (2025) for which no new data has been collected ($n = 1$) and for which an extended data record is now available ($n = 7$), and new sites for which no data has previously been presented ($n = 8$). Bureau of Meteorology seasonal rainfall – major zones is shown in the background (Government of Australia, 2025). In Australia, June – August is winter and December – February is summer.

2.2 Data analysis

All downloaded data were plotted and screened for false positive drips (e.g. disturbance by animals or external noise). Complete drip water and rainfall timeseries data sets for each site can be downloaded from the NGROS website (<https://groundwater.unsw.edu.au/>). As further quality control, potential recharge events were identified manually by examining periods with non-zero drip counts. Potential recharge events that occurred within 48 hours of each other were treated as one event for comparison to rainfall data.

The rainfall recharge threshold (RRT) defines the volume of rainfall required to exceed both evapotranspiration losses and sub-surface moisture deficits and thus initiate vertical water drainage (Baker et al., 2021). Thus, a site-dependent threshold, controlled by geology and antecedent climate conditions, can be calculated for each rainfall event by summing the rainfall leading to an observed onset of dripping. The thresholds for the individual events can then be summarized to understand rainfall amounts required for potential recharge initiation at a monthly, seasonal, or annual scale. For this study, rainfall thresholds corresponding to each event were calculated by plotting drip data and manually picking potential recharge events from the distinct, steep peaks in drip frequency at any logger that responded to a rainfall event. This study focusses on the timing, not the volume of potential recharge; thus, only the shape of the data, and not the magnitude of actual drip numbers was considered. Summation of the 48-hr rainfall totals (2 days) in advance of the onset of observed dripping provided an estimate of the RRT for each event. A 48-hr window was selected in Priestley et al. (2025) as it is large enough to capture the rainfall that causes a drip response and excludes subsequent rainfall events. Moreover, it aligns with the common water engineering period for rainfall and runoff accounting (Ball et al. 2019). The choice of antecedent rainfall window length is further mentioned in the Discussion.

3 Results

3.1 Climate and rainfall recharge thresholds

Average annual rainfall across all sites ranged from 477 – 1270 mm, with a median of 799 mm. Modelled potential evapotranspiration was 1261 – 2140 mm, with a median of 1488 mm. This resulted in aridity index values of 0.29 to 0.91 (median of 0.51); all sites are therefore rainfall limited (Figure 1). The number of potential recharge events captured at the sites ranged from 6 to 40 across 1 to 3 years of data (SI Table 1). Median rainfall recharge thresholds at each site ranged from 13.2 – 50.0 mm (mean of all sites = 26.8 mm for 48-hr antecedent rainfall). Across all sites, potential recharge occurred on 14 – 100% (median: 61%) of the days with 48-hr rainfall above the median RRT. Conversely, for all sites, potential recharge events also occurred on 3 – 20 occasions (median: 9 times over the monitoring period) when 48-hr rainfall was below the RRT. There was no statistical difference between rainfall recharge thresholds across 15 of the 16 sites, with pairwise Kruskal-Wallis with Dunn’s test for significance showing that of

the 78 site pairings, the RRT is only different for two pairings; Byaduk Lava Caves (BYA) had a lower rainfall recharge threshold than both WPM and YAR at 0.99 confidence.

3.2 Seasonality

Strong seasonality is apparent across the study sites, with the annual pattern of potential recharge skewed toward the season following the wet season and very little potential recharge in the season preceding the wet season (Figure 2). RRT thresholds are, on average, lower during the rainy season than during the dry season (Figure 2A). For the winter-dominated rainfall sites, the lower winter RRT occurs clearly in July and August, while there is great variability in the winter-dominated rainfall sites across most other months. Pairwise Kruskal-Wallis with Dunn's test shows the RRT is statistically different for two pairs of months at 0.95 confidence (January and July; February and July), and different at 0.99 confidence between January and July. The generally low level of difference is caused by the high RRT variability within each month. In the summer and autumn seasons, the monthly total P is similar to the 48-hr RRT, indicating that recharge is less likely to occur in these seasons (Figure 2B). This seasonal bias is also apparent in the number of potential recharge events (Figure 2C). Even though the RRT is only significantly different between January and July, the occurrence of potential recharge events for the winter rainfall-dominated and uniform rainfall sites is biased to winter and spring. 13.3% of all events occur in DJF, 12.5% in MAM; 42.1% in JJA; 32.1% in SON (Figure 2C).

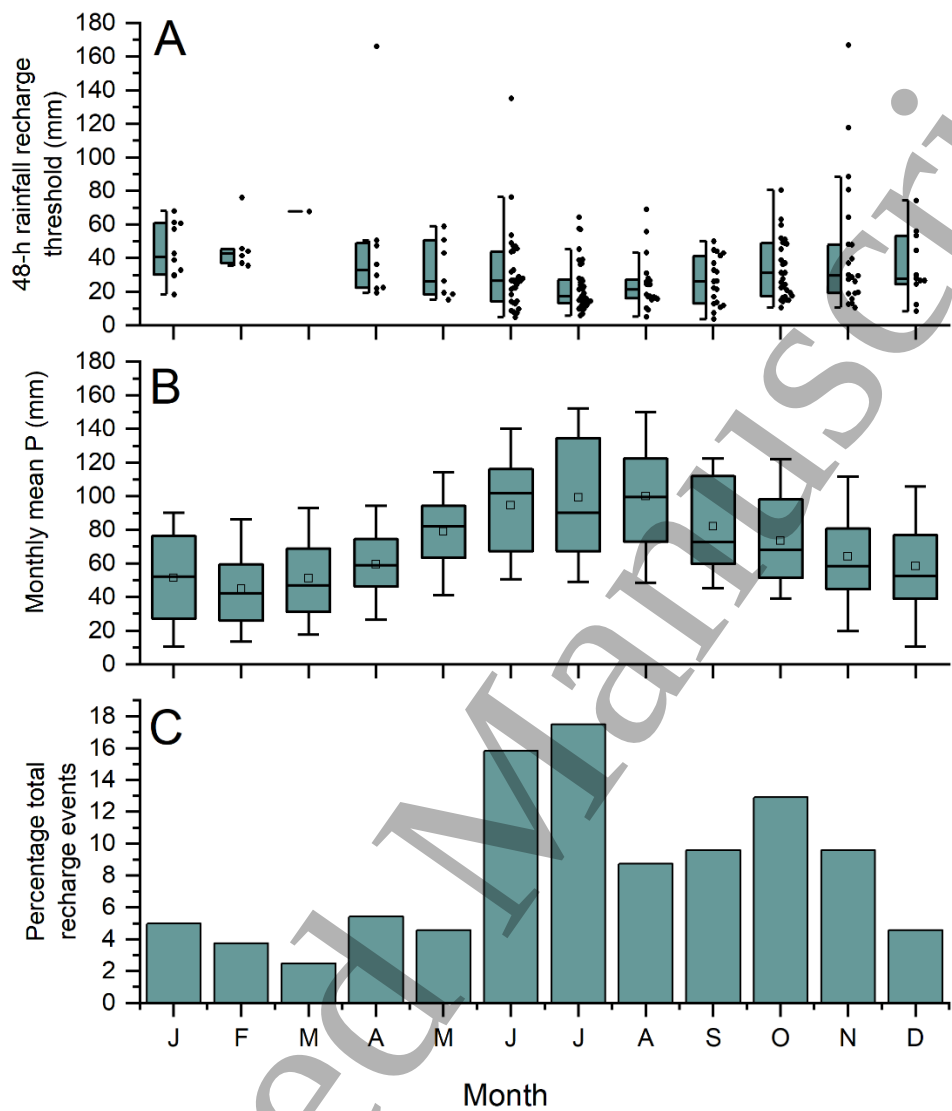


Figure 2. 48-hr rainfall recharge threshold (A), monthly mean precipitation (B), and percentage of all recorded potential recharge events (C) by month for winter-wet and uniform rainfall sites. June, July, and August are the winter months at the study sites.

3.3 Comparison of interannual recharge

Comparison of rainfall and potential recharge between sites shows commonalities across all sites but also regional distinctions. This is illustrated in the drip data; as an example, WAL and NAR show typical winter rainfall-dominated drip patterns that contrast sharply with YAR and CAP (Figure 3). At all sites time is needed to fill the soil moisture deficit with rainfall, and then subsequently relatively low rainfall events can lead to a drip event (i.e. onset of potential recharge). In fractured rock environments this has been described as a fill and spill type system (Guo et al., 2019). The impacts of a period of well below average monthly rainfall is apparent

across many of the sites in 2024-2025 with a reduced number of events compared to the previous year (Figure 3).

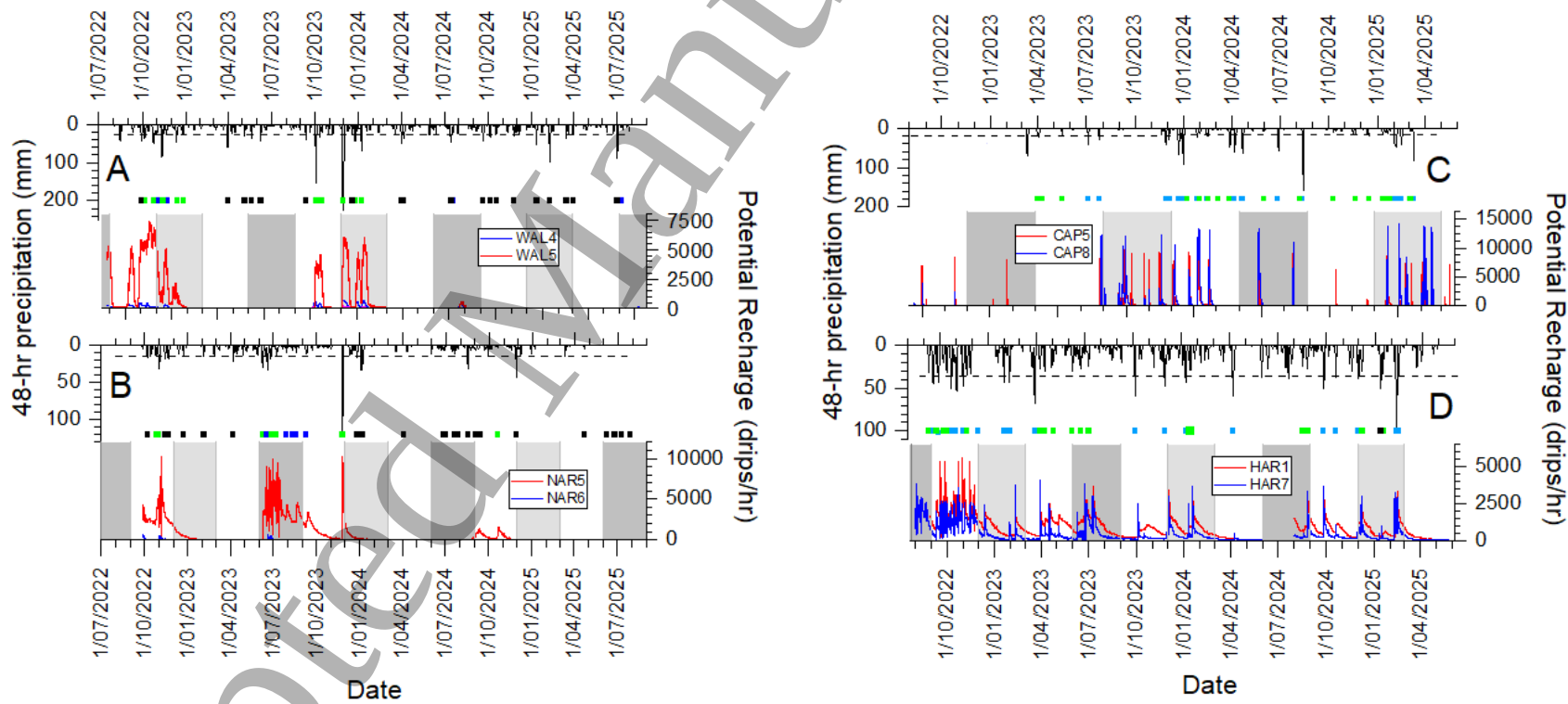
Time series data from several representative loggers at WAL shows that recharge has potentially occurred during the spring over the period of record (Figure 3A). Multiple daily (24-hr) rainfall events in the range of 30 – 60 mm (where the 48-hr median RRT is 26.8 mm) did not lead to potential recharge during autumn and winter in 2025. In contrast, in both 2023 and 2024, potential recharge events beginning in October (spring) persisted for several months, with drip rates renewed by periodic rain events above the RRT. For example, in 2023, high drip rates followed intense rainfall of approximately 150 mm over the preceding two days. In contrast, the comparatively dry autumn and winter of 2024 does not appear to have sufficiently wetted the subsurface (despite several rain events greater than 50 mm), and only about a week of recharge potentially occurred that year, despite recurring but relatively low volume (approx. 10–25 mm over 48-72 hr) rainfall. The impact of drought in 2025 is readily apparent; the final two events at Walhalla had relatively low 48-hr rainfall, but over 80 mm of precipitation in the week before each event. This suggests a longer time for this site is needed to wet the unsaturated zone during dry periods and hence ‘set the site up’ for recharge events to occur.

Rainfall is sparser and has lower volume at NAR (Figure 3B). Rainfall is primarily confined to winter months at NAR, where recharge potentially began on 8 June 2023 following a cumulative volume of 81 mm of rainfall spread out relatively evenly over the preceding 40 days. Dripping persisted for over six months. However, in 2024 recharge potentially occurred only for a matter of days, beginning on 17 August. The site had received a cumulative 131 mm of rainfall during that spring and summer, but the rainfall volumes were consistently low (the highest 24-hr volume was under 12 mm) and discontinuous. Rainfall remained sparse through the first half of 2025 and no drips were recorded; this was one of the driest years on record in southeast South Australia.

In stark contrast to these isolated recharge patterns, the recharge at YAR and CAP is potentially more frequent and immediately responsive to rainfall (Figure 3C and D). Both sites are in mature karst geology, where solutional widened fractures and caves control the flow paths. At CAP, rainfall intensity is typically high (> 20 mm in 24 hr), and reliably leads to intense, short duration groundwater potential recharge. In contrast, at the mountainous YAR site, dripping persists throughout the year, with drip rates responding rapidly to repeated high intensity rainfall events throughout the year. The median RRT at these sites does not appear to predict potential recharge well; potential recharge occurs often when precipitation is below the threshold.

At both YAR and CAP, there is sparse vegetation and only a thin soil horizon; thus, evapotranspiration and accumulated soil moisture deficit would be expected to exhibit little influence over recharge. In contrast, both sites show evidence of the spill and fill hydrology typical of karst regions (Figure 3C and D). Once the median annual RRT is reached at these sites, potential recharge always occurs at these sites.

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249

250 **Figure 3. Time series of precipitation and potential recharge events recorded at selected loggers at (A) Walhalla Mine, (B)**
251 **Naracoorte Caves, (C) Capricorn Caves, and (D) Yarrangobilly Harrie Wood Cave. For each site, blue dots show 48-hr rainfall above**
252 **the rainfall recharge threshold (RRT) and potential recharge occurred, green show potential recharge with rainfall under the**
253 **threshold, and black show rainfall above the threshold but no recharge occurred. Dark grey shaded areas indicate winter (June-**
254 **August) and light grey indicates summer (December – February) seasons. Note that CAP has summer dominant rainfall, while the**
255 **others are winter rainfall sites.**

4 Discussion

Groundwater recharge is typically modelled and managed using annual estimates; however, at an annual scale, we found no statistical difference between median RRTs at 75 out of 78 unique site-pairs (Kruskal-Wallis ANOVA with Dunn's test at 95% confidence), despite marked differences in climate and geology. Yet, we see that recharge thresholds clearly vary seasonally across all seasonally dominated rainfall sites (Fig. 2). Moreover, rainfall and potential recharge patterns at studied winter-wet sites clearly show that little to no recharge occurs in late summer / early autumn (Fig. 3). Therefore, accurate estimation of groundwater recharge for management of water resources should focus on estimating groundwater recharge from wet season rainfall events.

Recharge is typically understood to correlate with rainfall, with global models often estimating recharge as simply a percentage of rainfall, moderated by factors such as geology and soil type (e.g. Döll et al., 2002). Using this simple relationship, we could use our median RRTs to extrapolate the number of anticipated annual days of potential groundwater recharge from annual maps of annual rainfall event recurrence; for example, the 10 mm and 25 mm rainfall event maps available online through the Australian Bureau of Meteorology (BoM, 2025b). This may be an accurate approach at some sites; indeed Gnann et al. (2023) suggested that globally there is little variation in recharge with precipitation at relatively low annual rainfall volumes. However, recent research shows that system response to climatic perturbations in climate-sensitive regions is dependent on "hydrological memory" stored within the porous medium (Pan et al., 2025). Our results concur and we caution against a blanket determination of recharge based on annual precipitation or rainfall volume and instead demonstrate the importance of antecedent conditions in deciding whether rain events of a given volume over a specified time period will result in recharge (Fig 5 and 6).

Rainfall in Australia is known to be highly variable in time and space, regionally controlled by the interacting influences of the El Niño–Southern Oscillation (ENSO), Indian Ocean dipole (IOD), and the southern annular mode (SAM; also known as the Antarctic Oscillation) (King et al., 2014). The regionally wet phase of ENSO, the La Niña, leads to above-average winter and spring rainfall across the north and east of Australia, seasons where our evidence suggests recharge is most likely. Both ENSO and SAM patterns are impacted by climate change, making it difficult to predict future rainfall patterns, although the SAM is likely to continue its positive trend, with decreasing winter rainfall along the southern coast of the continent (Purich et al., 2025; Taschetto et al., 2025) suggesting that the RRT will be exceeded less often in that region. Our results show that shifts in rainfall timing and volume due to changes in these atmospheric rainfall drivers resulted in significant inter-year variability in recharge. According to the Australian Bureau of Meteorology, repeated high pressure weather patterns throughout 2024 and the first half of 2025 led to persistent warm and dry conditions (BoM, 2025a). These conditions were also consistent with a long-term declining trend in rainfall during the cool season months from April to October across southern Australia due to a positive southern annular mode (SAM) (Commonwealth of Australia, 2024). This evidence all suggests that the recurrence of recharge events will decrease along the southern coast of Australia.

The results of this study clearly reflect the seasonal cycle of groundwater recharge (Fig. 4). Seasonality of recharge has been discussed previously in the literature. However, previous

studies have typically used stable isotopes (e.g. Liu et al., 2021; Yeh et al., 2014), which are useful for determining overall seasonality in the recharge signature but cannot resolve event-scale response. Our study results help to clearly distinguish processes occurring throughout the year to build a conceptual model of recharge response to rainfall events. In winter rainfall-dominated regions, consistent, low-volume, early-mid season rainfall fills up soil storage and shallow fractures. Once these are filled, subsequent rainfall events at or even below the annual median RRT will potentially result in groundwater recharge. These conditions are typically maintained through the beginning of spring, resulting in the potential for spring recharge events at relatively low rainfall amounts. However, by summer, the soil conditions are dry and evapotranspiration is high, and even high-volume summer or autumn rainfall is unlikely to result in groundwater recharge.

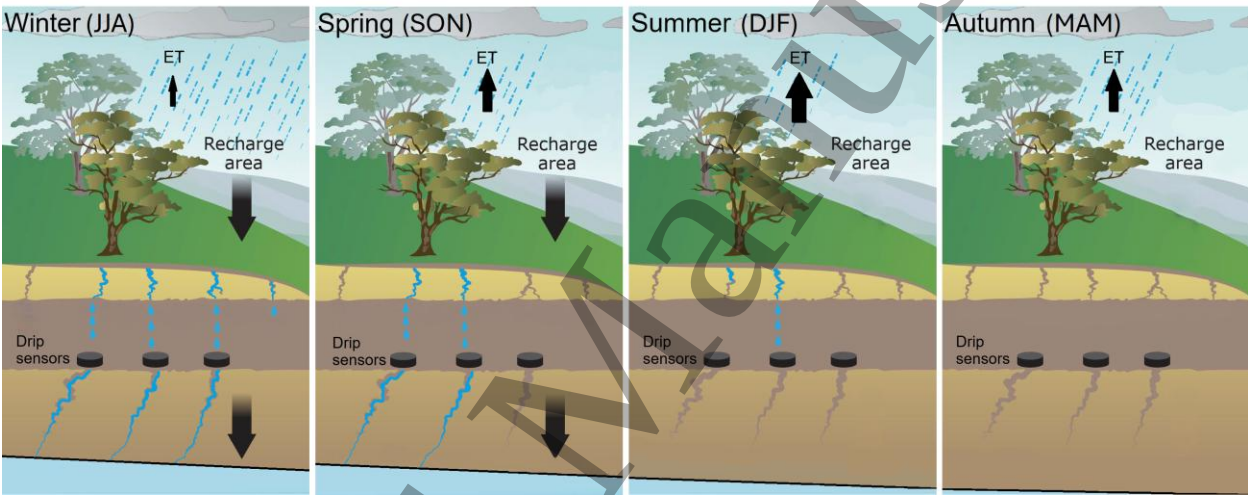


Figure 4. Conceptual model of the seasonal progression of recharge at winter-rainfall sites. The initials of the months corresponding to each season are listed at the top of each panel. Background figure produced by Anna Blacka (Water Research Laboratory, UNSW).

In addition to rainfall, other factors such as soil cover, evapotranspiration, and rainfall intensity are now understood to exert strong controls over groundwater recharge, blurring the relationship between rainfall and groundwater recharge (Qiu et al., 2025; Wang et al., 2015). At another Australian cave, Song et al. (2025) suggested the dominance of soil cover in determining RRTs. In our study, the sparse soil cover at sites such as YAR and CAP likely contributes to the reliable recharge response to rainfall events all year. Evapotranspiration is the main water balance process occurring all year and leading to a soil water deficit; the aridity index is below 1 at all sites (Table SI). Because of this, the RRT is high and recharge is of short duration at most sites, and high-volume rain events are needed even during the wet season for potential recharge to occur. In contrast, similar volume rain events during the dry season do not lead to recharge. This is because late summer 48-hr rainfall recharge thresholds are roughly equivalent to monthly average rainfall. Similar seasonality of recharge has been noted in cold, humid winter-wet climates in the northern hemisphere, with modelling results suggesting that the seasonality of recharge will become even more pronounced under climate change (Dubois et al., 2022). For example, despite the decreasing rainfall trends overall, the return interval for

extreme rainfall events in much of Australia is increasing (Gallant et al., 2014; Wasko et al., 2024). In a global study of seasonality in recharge using stable isotope data, Jasechko et al. (2014) also concluded that in temperate regions, winter precipitation is better able to produce recharge than summer rainfall, although the figures in that study indicated little seasonal change in isotopic values at Australian sites. In contrast to that study, we found very pronounced seasonal differences at all winter-wet sites. We recommend using a rainfall seasonality classification in preference to a climate classification for predicting the seasonality of groundwater recharge.

Despite the availability of increased spatial and temporal coverage of drip sites used in this study, several knowledge gaps remain. Our network is focused on fractured rock regions, predominantly in the southeast region of the country, where access to underground sites is available, and our results need to be compared to recharge in porous media such as unconsolidated sands and alluvial sediments. The importance of evapotranspiration and soil moisture is indirectly apparent in these results, and future research including these parameters at daily resolution is important for understanding the physical mechanisms controlling when and where recharge events occur. Moreover, our study period reported here is marked by drought conditions across multiple sites. The study period captured both El Nino and La Nino years and was therefore largely representative of typical climate variability in the study region. However, longer term data would be needed to explore rainfall recharge thresholds across various climate states. Long-term monitoring continues, which we envisage will enable us to better understand the rainfall – soil moisture – groundwater relationship at the event scale throughout drought, normal, and wet years and better understand how those relationships may be impacted by climate change.

5 Conclusions

Although there are several studies on groundwater recharge at the Australian continental scale (e.g. Crosbie et al., 2013; Lee et al., 2024), they have focused on long-term patterns or future projections; this study represents the first time that event-scale potential groundwater recharge thresholds have been examined at a continental scale using fine-scale field-data, allowing conclusions on seasonal recharge variation. We show that median RRTs are generally statistically indistinguishable across Australia; in contrast, seasonality is important in predicting recharge. At our winter-wet sites, there is often a complex relationship between individual rainfall events and whether recharge occurred or not. Dry years result in a lack of recharge even at rainfall volumes above the predicted threshold, showing the role of antecedent moisture in winter-wet regions and the role of ET drying down the soil profile. Evidence in this study indicates that RRT are variable over time, and thus periods of drought and climate change are critical considerations for groundwater recharge and availability. This study adds a broader understanding of rainfall recharge thresholds and their drivers across a wide swath of Australia, with new insights on the role of seasonality in influencing the RRT and the reliability of recharge occurring during the wet season. It is evident that multiple years of data are needed to allow for interannual variability and the revision of RRT values. This will facilitate a clearer understanding of how the threshold changes seasonally and in response to climate extremes, such as droughts.

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Open Research

The data used in this manuscript is available from the NGROS database at <https://groundwater.unsw.edu.au>.

Conflict of Interest Disclosure

The authors declare there are no conflicts of interest for this manuscript.

Ethics

No ethics approval was required for this study.

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